

# **ACADEMIC REGULATIONS & SYLLABUS**

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**Faculty of Pharmacy  
Master of Pharmacy Programme  
(Pharmacy Practice)**

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**(AS PER PCI SYLLABUS)**

**A. Y. 2023 - 2024**

**Ramanbhai Patel College of Pharmacy  
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**

# **CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**

## **MASTER OF PHARMACY (M. Pharm.) PROGRAMME**

### **Vision of RPCP**

*To Become a Premier Pharma Institute by Creating World Class Pharmacists  
and Researchers.*

### **Mission of RPCP**

*To Strive for the Excellence in Pharmaceutical Sciences through Quality  
Education and Research.*

## PROGRAM OUTCOMES

- 1. Pharmacy Knowledge:** Possess knowledge and comprehension of the core and basic knowledge associated with the profession of pharmacy, including biomedical sciences; pharmaceutical sciences; behavioral, social, and administrative pharmacy sciences; and manufacturing practices.
- 2. Planning Abilities:** Demonstrate effective planning abilities including time management, resource management, delegation skills and organizational skills. Develop and implement plans and organize work to meet deadlines.
- 3. Problem analysis:** Utilize the principles of scientific enquiry, thinking analytically, clearly and critically, while solving problems and making decisions during daily practice. Find, analyze, evaluate and apply information systematically and shall make defensible decisions.
- 4. Modern tool usage:** Learn, select, and apply appropriate methods and procedures, resources, and modern pharmacy-related computing tools with an understanding of the limitations.
- 5. Leadership skills:** Understand and consider the human reaction to change, motivation issues, leadership and team-building when planning changes required for fulfillment of practice, professional and societal responsibilities. Assume participatory roles as responsible citizens or leadership roles when appropriate to facilitate improvement in health and wellbeing.
- 6. Professional Identity:** Understand, analyze and communicate the value of their professional roles in society (e.g. health care professionals, promoters of health, educators, managers, employers, employees).
- 7. Pharmaceutical Ethics:** Honour personal values and apply ethical principles in professional and social contexts. Demonstrate behavior that recognizes cultural and personal variability in values, communication and lifestyles. Use ethical frameworks; apply ethical principles while making decisions and take responsibility for the outcomes associated with the decisions.
- 8. Communication:** Communicate effectively with the pharmacy community and with society at large, such as, being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
- 9. The Pharmacist and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and the consequent responsibilities relevant to the professional pharmacy practice.
- 10. Environment and sustainability:** Understand the impact of the professional pharmacy solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 11. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. Self-assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

**FACULTY OF PHARMACY**  
**ACADEMIC REGULATIONS**  
**MASTER OF PHARMACY (M. Pharm.) PROGRAMME**  
**Choice Based Credit System (CBCS)**

**1. Short Title and Commencement**

These regulations shall be called as “The Revised Academic Regulations for the postgraduate programmes under the Faculty of Pharmacy”. They shall come into effect from the Academic Year 2018-19. The regulations framed are subject to modifications from time to time by the respective regulatory bodies.

**2. Minimum Qualification for Admission**

2.1 Candidate shall have passed B. Pharm Degree examination of an Indian university established by law in India from an institution approved by Pharmacy Council of India and has scored not less than 55 % of the maximum marks (aggregate of 4 years of B.Pharm)

2.2 Every student, selected for admission to post graduate pharmacy program in any PCI approved institution should have obtained registration with the State Pharmacy Council or should obtain the same within one month from the date of his/her admission, failing which the admission of the candidate shall be cancelled.

Note: It is mandatory to submit a migration certificate obtained from the respective university where the candidate had passed his/her qualifying degree (B.Pharm).

**3. Duration of the Programme**

The course of study for M.Pharm shall extend over a period of four semesters (two academic years). The curricula and syllabi for the program shall be prescribed from time to time by Pharmacy Council of India, New Delhi.

**4. Medium of Instruction and Examinations**

Medium of instruction and examination shall be in English.

**5. Working Days in a Semester**

Each semester shall consist of not less than 100 working days. The odd semesters shall be conducted from the month of June/July to November/December and the even semesters shall be conducted from December/January to May/June in every calendar year.

## **6. Attendance and Progress**

A candidate is required to put in at least 80% attendance in individual courses considering theory and practical separately. The candidate shall complete the prescribed course satisfactorily to be eligible to appear for the respective examinations.

## **7. Programme Credit Structure**

As per the philosophy of Credit Based Semester System, certain quantum of academic work viz. theory classes, tutorial hours, practical classes, etc. are measured in terms of credits. On satisfactory completion of the courses, a candidate earns credits. The amount of credit associated with a course is dependent upon the number of hours of instruction per week in that course. Similarly, the credit associated with any of the other academic, co/extra-curricular activities is dependent upon the quantum of work expected to be put in for each of these activities per week.

### **7.1. Credit assignment**

#### **7.1.1. Theory and Laboratory courses**

Courses are broadly classified as Theory and Practical. Theory courses consist of lecture (L) and Practical (P) courses consist of hours spent in the laboratory. Credits (C) for a course is dependent on the number of hours of instruction per week in that course, and is obtained by using a multiplier of one (1) for lecture and a multiplier of half (1/2) for practical (laboratory) hours. Thus, for example, a theory course having four lectures per week throughout the semester carries a credit of 4. Similarly, a practical having four laboratory hours per week throughout semester carries a credit of 2.

The contact hours of seminars, assignments and research work shall be treated as that of practical courses for the purpose of calculating credits. i.e., the contact hours shall be multiplied by 1/2. Similarly, the contact hours of journal club, research work presentations and discussions with the supervisor shall be considered as theory course and multiplied by 1.

### **7.2. Minimum credit requirements**

The minimum credit points required for the award of M. Pharm. degree is 93. However based on the credit points earned by the students under the head of co-curricular activities, a student shall earn a maximum of 98 credit points. These credits are divided into Theory courses, Practical, Seminars, Assignments, Research work, Discussions with the supervisor, Journal club and Co-Curricular activities over the duration of four semesters. The credits are distributed semester-wise. Courses generally progress in sequence, building competencies and their positioning indicates

certain academic maturity on the part of the learners. Learners are expected to follow the semester-wise schedule of courses given in the syllabus.

## 8. Academic work

A regular record of attendance both in Theory, Practical, Seminar, Assignment, Journal club, Discussion with the supervisor, Research work presentation and Dissertation shall be maintained by the department / teaching staff of respective courses.

## 9. Examinations/Assessments

The scheme for internal assessment and end semester examinations is given in Annexure II (Table – 1 to 5).

### 9.1. End Semester Examinations

The End Semester Examinations for each theory and practical course through semesters I to IV shall be conducted by the University for which Examinations shall be conducted by the subject experts at college level and the marks/grades shall be submitted to the university.

### 9.2. Internal Assessment: Continuous Mode

The marks allocated for Continuous mode of Internal Assessment shall be awarded as per the scheme given below.

**Table- 1: Scheme for awarding internal assessment: Continuous mode**

| <b>Theory</b>                                       |                      |
|-----------------------------------------------------|----------------------|
| <b>Criteria</b>                                     | <b>Maximum Marks</b> |
| Attendance (Refer Table – 2)                        | 8                    |
| Student – Teacher interaction                       | 2                    |
| <b>Total</b>                                        | 10                   |
| <b>Practical</b>                                    |                      |
| Attendance (Refer Table – 2)                        | 10                   |
| Based on Practical Records, Regular viva voce, etc. | 10                   |
| <b>Total</b>                                        | 20                   |

**Table- 2: Guidelines for the allotment of marks for attendance**

| <b>Percentage of Attendance</b> | <b>Theory</b> | <b>Practical</b> |
|---------------------------------|---------------|------------------|
| 95 – 100                        | 8             | 10               |
| 90 – 94                         | 6             | 7.5              |
| 85 – 89                         | 4             | 5                |
| 80 – 84                         | 2             | 2.5              |
| Less than 80                    | 0             | 0                |

### 9.2.1. Sessional Exams

Two Sessional exams shall be conducted for each theory / practical course as per the schedule fixed by the college. The scheme of question paper for theory and practical Sessional examinations will be as prescribed by the regulatory body. The average marks of two Sessional exams shall be computed for internal assessment as per the requirements given in Annexure II. Sessional exam shall be conducted for 30 marks for theory and shall be computed for 15 marks. Similarly, Sessional exam for practical shall be conducted for 40 marks and shall be computed for 10 marks.

### 10. Promotion and Award of Grades

A student shall be declared PASS and eligible for getting grade in a course of M.Pharm. Programme, if he/she secures at least 50% marks in that particular course including internal assessment.

### 11. Carry Forward of Marks

In case a student fails to secure the minimum 50% in any Theory or Practical course as specified in 9 above, then he/she shall reappear for the end semester examination of that course. However, his/her marks of the Internal Assessment shall be carried over and he/she shall be entitled for grade obtained by him/her on passing.

### 12. Improvement of Internal Assessment

A student shall have the opportunity to improve his/her performance only once in the Sessional exam component of the internal assessment. The re-conduct of the Sessional exam shall be completed before the commencement of next end semester theory examinations.

### 13. Re-examination of End Semester Examinations

Re-examination of end semester examination shall be conducted as per the schedule given in table 3. The exact dates of examinations shall be notified from time to time.

**Table-3: Tentative schedule of end semester examinations**

| Semester  | For Regular Candidates | For Failed Candidates                                                          |                         |
|-----------|------------------------|--------------------------------------------------------------------------------|-------------------------|
|           |                        | Remedial Examination-1                                                         | Remedial Examination-2* |
| I and III | November / December    | After 15 days of the declaration of the result of the end semester Examination | last week of June/      |

|           |            |                                                                                |                                            |
|-----------|------------|--------------------------------------------------------------------------------|--------------------------------------------|
| II and IV | May / June | After 15 days of the declaration of the result of the end semester Examination | end of the Academic Year, which is earlier |
|-----------|------------|--------------------------------------------------------------------------------|--------------------------------------------|

\* If student who are remain left from end semester examination and remedial examination-1.

#### 14. Academic Progression

No student shall be admitted to any examination unless he/she fulfils the norms given in item no. 6 under the heading of attendance and progress. Academic progression rules are applicable as follows:

- “Student will not be allowed to move to next year if she/he has not cleared all the courses of the previous year”
- A student shall be eligible to carry forward all the courses of I<sup>st</sup> semesters till the II<sup>nd</sup> semester examinations. However, he/she shall not be eligible to attend the courses of the III<sup>rd</sup> semester until all the courses of the I<sup>st</sup> and II<sup>nd</sup> semesters are successfully completed.
- A student shall be eligible to get his/her CGPA upon successful completion of the courses of I to IV semesters within the stipulated time period as per the norms.
- Note: Grade “NA” should be considered as failed and treated as one head for deciding ATKT. Such rules are also applicable for those students who fail to register for examination(s) of any course in any semester.

#### 15. Grading of Performances (Letter Grades and Grade Points Allocations)

Based on the performances, each student shall be awarded a final letter grade at the end of the semester for each course. The letter grades and their corresponding grade points are given in Table – 4.

**Table-4: Letter grades and grade points equivalent to Percentage of marks and performances**

| Percentage of Marks Obtained | Letter Grade | Grade Point | Performance |
|------------------------------|--------------|-------------|-------------|
| 90.00 – 100                  | AA           | 10          | Outstanding |
| 80.00 – 89.99                | AB           | 9           | Excellent   |
| 70.00 – 79.99                | BB           | 8           | Good        |
| 60.00 – 69.99                | BC           | 7           | Fair        |
| 50.00 – 59.99                | CC           | 6           | Average     |
| Less than 50                 | FF           | 0           | Fail        |
| Absent                       | NA           | 0           | Fail        |



A learner who remains absent for any end semester examination shall be assigned a letter grade of “NA” and a corresponding grade point of zero. He/she should reappear for the said evaluation/examination in due course.

### **16. Semester Grade Point Average (SGPA)**

The performance of a student in a semester is indicated by a number called ‘Semester Grade Point Average’ (SGPA). The SGPA is the weighted average of the grade points obtained in all the courses by the student during the semester. For example, if a student takes five courses(Theory/Practical) in a semester with credits C1, C2, C3, C4 and C5 and the student’s grade points in these courses are G1, G2, G3, G4 and G5, respectively, and then students’ SGPA is equal to:

$$SGPA = \frac{C1G1+C2G2+C3G3+C4G4}{C1+C2+C3+C4}$$

The SGPA is calculated to two decimal points. It should be noted that, the SGPA for any semester shall take into consideration the FF and Ab. grade awarded in that semester. For example if a learner has a FF or Ab. grade in course 4, the SGPA shall then be computed as:

$$SGPA = \frac{C1G1+C2G2+C3G3+C4Zero}{C1+C2+C3+C4}$$

### **17. Cumulative Grade Point Average (CGPA)**

The CGPA is calculated with the SGPA of all the VIII semesters to two decimal points and is indicated in final grade report card/final transcript showing the grades of all VIII semesters and their courses. The CGPA shall reflect the failed status in case of FF grade(s), till the course(s) is/are passed. When the course(s) is/are passed by obtaining a pass grade on subsequent examination(s) the CGPA shall only reflect the new grade and not the fail grades earned earlier. The CGPA is calculated as:

$$CGPA = \frac{C1S1+C2S2+C3S3+C4S4}{C1+C2+C3+C4}$$

where C1, C2, C3,... is the total number of credits for semester I,II,III,... and S1,S2, S3,...is the SGPA of semester I,II,III,.... .

**No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.**

### 18. Declaration of Class (Table-5)

The class shall be awarded on the basis of CGPA as follows:

|                              |                         |
|------------------------------|-------------------------|
| First Class with Distinction | = CGPA of. 7.50 to 10.0 |
| First Class                  | = CGPA of 6.0 to 7.49   |
| Second Class                 | = CGPA of 5.0 to 5.99   |
| Pass Class                   | < CGPA of 5.00          |

### 19. Project Work

All the students shall undertake a project under the supervision of a teacher in Semester III to IV and submit a report. 4 copies of the project report shall be submitted (typed & bound copy not less than 75 pages).

The internal and external examiner appointed by the University shall evaluate the project at the time of the Practical examinations of other semester(s). The projects shall be evaluated as per the criteria given below.

| Evaluation of Dissertation Book |                             |                             |                             |                             |
|---------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Criteria                        | Semester-III                |                             | Semester-IV                 |                             |
|                                 |                             |                             | Internal Evaluation (Marks) | External Evaluation (Marks) |
| Objective(s) of the work done   | --                          |                             | 05                          | 05                          |
| Methodology Adopted             | --                          |                             | 25                          | 25                          |
| Results and Discussions         | --                          |                             | 15                          | 15                          |
| Conclusions and Outcome         | --                          |                             | 05                          | 05                          |
| Total                           | --                          |                             | 50                          | 50                          |
| Final Total                     |                             |                             | 100                         |                             |
| Evaluation of Presentation      |                             |                             |                             |                             |
|                                 | Semester-III                |                             | Semester-IV                 |                             |
| Criteria                        | Internal Evaluation (Marks) | External Evaluation (Marks) | Internal Evaluation (Marks) | External Evaluation (Marks) |
| Presentation of work            | 75                          | 100                         | 75                          | 75                          |
| Communication skills            | 25                          | 50                          | 25                          | 25                          |
| Question and answer skills      | 50                          | 50                          | 50                          | 50                          |
| Total                           | 150                         | 200                         | 150                         | 150                         |
| Final Total                     | 350                         |                             | 300                         |                             |

## 20. Award of Ranks

Ranks and Medals shall be awarded on the basis of final CGPA. However, candidates who fail in one or more courses during the M.Pharm program shall not be eligible for award of ranks. Moreover, the candidates should have completed the M. Pharm program in minimum prescribed number of years, (two years) for the award of Ranks.

## 22. Award of degree

Candidates who fulfil the requirements mentioned above shall be eligible for award of degree during the ensuing convocation.

## 23. Duration for completion of the program of study

The duration for the completion of the program shall be fixed as double the actual duration of the program and the students have to pass within the said period, otherwise they have to get fresh Registration.

## 24. Extra Credit:

An extra credit is to be offered to a student for achievements in co-curricular and extra-curricular activities. This credit shall not be counted while considering the minimum credits for completing the program. The activities and appropriate weight (points) to be allocated to award an extra credit are broadly classified as per the table below:

| Sr. no. | Name of the Activity                                                                                                                         | Maximum Credit Points Eligible / Activity |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| 1       | Participation in National Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student)      | 01                                        |
| 2       | Participation in international Level Seminar/Conference/Workshop/Symposium/ Training Programs (related to the specialization of the student) | 02                                        |
| 3       | Academic Award/Research Award from State Level/National Agencies                                                                             | 01                                        |
| 4       | Academic Award/Research Award from International Agencies                                                                                    | 02                                        |
| 5       | Research / Review Publication in National Journals (Indexed in Scopus / Web of Science)                                                      | 01                                        |
| 6       | Research / Review Publication in International Journals (Indexed in Scopus / Web of Science)                                                 | 02                                        |

**Note: International Conference: Held Outside India**

**International Journal: The Editorial Board Outside India**

**\*The credit points assigned for extracurricular and or co-curricular activities shall be given by the Principals of the colleges and the same shall be submitted to the University. The criteria to acquire this credit point shall be defined by the colleges from time to time.**

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**MASTER OF PHARMACY (PHARMACY PRACTICE) PROGRAMME**  
**Schemes for internal assessments and end semester examinations**

**SEMESTER-I**

**SCHEME OF TEACHING**

| Course Code        | Course Name                            | Credit Hours | Credit Points | Hrs/Week  |
|--------------------|----------------------------------------|--------------|---------------|-----------|
| PHPPP001           | Clinical Pharmacy Practice             | 4            | 4             | 4         |
| PHPPP002           | Pharmacotherapeutics-I                 | 4            | 4             | 4         |
| PHPPP003           | Hospital & Community Pharmacy          | 4            | 4             | 4         |
| PHPPP014           | Clinical Research                      | 4            | 4             | 4         |
| PHPPP005           | Pharmacy Practice Practical-I          | 12           | 6             | 12        |
| PHPPP006           | Seminar/Assignment- I (Practical mode) | 2            | 1             | 2         |
| ---                | DHSS Elective-I*                       | 2            | 2             | 2         |
| <b>Total (118)</b> |                                        | <b>32</b>    | <b>25</b>     | <b>32</b> |

**SCHEME OF EVALUATION**

| Course Code | Course                                 | Internal Assessment |                 |          |       | End Semester Exams |          | Total Marks |
|-------------|----------------------------------------|---------------------|-----------------|----------|-------|--------------------|----------|-------------|
|             |                                        | Continuo<br>us Mode | Sessional Exams |          | Total | Marks              | Duration |             |
|             |                                        |                     | Marks           | Duration |       |                    |          |             |
| PHPPP001    | Clinical Pharmacy Practice             | 10                  | 15              | 1 Hr     | 25    | 75                 | 3 Hrs    | 100         |
| PHPPP002    | Pharmacotherapeutics-I                 | 10                  | 15              | 1 Hr     | 25    | 75                 | 3 Hrs    | 100         |
| PHPPP003    | Hospital & Community Pharmacy          | 10                  | 15              | 1 Hr     | 25    | 75                 | 3 Hrs    | 100         |
| PHPPP014    | Clinical Research                      | 10                  | 15              | 1 Hr     | 25    | 75                 | 3 Hrs    | 100         |
| PHPPP005    | Pharmacy Practice Practical-I          | 20                  | 30              | 6 Hrs    | 50    | 100                | 6 Hrs    | 150         |
| PHPPP006    | Seminar/Assignment- I (Practical mode) | -                   | -               | -        | -     | -                  | -        | 100         |
| ---         | DHSS Elective-I*                       | -                   | -               | -        | 30    | 70                 | -        | 100         |
| Total       |                                        |                     |                 |          |       |                    |          | 750         |

**\*DHSS elective courses: SCHEME OF TEACHING**

| Semester | Course Code | Course Name                               | Credit Hours | Credit Points | Hrs/Week |
|----------|-------------|-------------------------------------------|--------------|---------------|----------|
| I        | HS105.02 B  | Academic Speaking and Presentation Skills | 02           | 02            | 02       |

**\*DHSS elective courses: SCHEME OF EVALUATION**

| Course Code | Course Name                               | Evaluation Scheme |          |           |          |       |
|-------------|-------------------------------------------|-------------------|----------|-----------|----------|-------|
|             |                                           | Theory            |          | Practical |          | Total |
|             |                                           | Internal          | External | Internal  | External |       |
| HS105.02 B  | Academic Speaking and Presentation Skills | -                 | -        | 30        | 70       | 100   |

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**MASTER OF PHARMACY (PHARMACY PRACTICE) PROGRAMME**  
**Schemes for internal assessments and end semester examinations**  
**SEMESTER-II**

**SCHEME OF TEACHING**

| Course Code  | Course Name                                               | Credit Hours | Credit Points | Hrs/Week  |
|--------------|-----------------------------------------------------------|--------------|---------------|-----------|
| PHPPP015     | Principles of Quality Use of Medicines                    | 4            | 4             | 4         |
| PHPPP007     | Pharmacotherapeutics II                                   | 4            | 4             | 4         |
| PHPPP008     | Clinical Pharmacokinetics and Therapeutic Drug Monitoring | 4            | 4             | 4         |
| PHPPP009     | Pharmacoepidemiology & Pharmacoeconomics                  | 4            | 4             | 4         |
| PHPPP010     | Pharmacy Practice Practical II                            | 12           | 6             | 12        |
| PHPPP011     | Seminar/Assignment-II                                     | 2            | 1             | 2         |
| ---          | DHSS elective-II*                                         | 2            | 2             | 2         |
| ---          | University Elective-II**                                  | 2            | 2             | 2         |
| <b>Total</b> |                                                           | <b>34</b>    | <b>27</b>     | <b>34</b> |

**SCHEME OF EVALUATION**

| Course Code | Course                                                    | Internal Assessment |                 |              |       | End Semester Exams |              | Total Marks |
|-------------|-----------------------------------------------------------|---------------------|-----------------|--------------|-------|--------------------|--------------|-------------|
|             |                                                           | Continuou<br>s Mode | Sessional Exams |              | Total | Marks              | Duratio<br>n |             |
|             |                                                           |                     | Marks           | Duratio<br>n |       |                    |              |             |
| PHPPP015    | Principles of Quality Use of Medicines                    | 10                  | 15              | 1Hr          | 25    | 75                 | 3Hrs         | 100         |
| PHPPP007    | Pharmacotherapeutics II                                   | 10                  | 15              | 1Hr          | 25    | 75                 | 3Hrs         | 100         |
| PHPPP008    | Clinical Pharmacokinetics and Therapeutic Drug Monitoring | 10                  | 15              | 1Hr          | 25    | 75                 | 3Hrs         | 100         |
| PHPPP009    | Pharmacoepidemiology & Pharmacoeconomics                  | 10                  | 15              | 1Hr          | 25    | 75                 | 3Hrs         | 100         |
| PHPPP010    | Pharmacy Practice Practical II                            | 20                  | 30              | 6Hrs         | 50    | 100                | 6Hrs         | 150         |
| PHPPP011    | Seminar/Assignment-II                                     | -                   | -               | -            | -     | -                  | -            | 100         |
| ---         | DHSS elective-II*                                         | -                   | -               | -            | 30    | 70                 | -            | 100         |
| ---         | University Elective-II**                                  | -                   | -               | -            | 30    | 70                 | -            | 100         |
| Total       |                                                           |                     |                 |              |       |                    |              | 850         |

**\*DHSS elective courses: SCHEME OF TEACHING**

| Semester | Course Code | Course Name      | Credit Hours | Credit Points | Hrs/Week |
|----------|-------------|------------------|--------------|---------------|----------|
| II       | HS106.02 B  | Academic Writing | 02           | 02            | 02       |

**\*DHSS elective courses: SCHEME OF EVALUATION**

| Course Code | Course Name      | Evaluation Scheme |          |           |          |       |
|-------------|------------------|-------------------|----------|-----------|----------|-------|
|             |                  | Theory            |          | Practical |          | Total |
|             |                  | Internal          | External | Internal  | External |       |
| HS106.02 B  | Academic Writing | -                 | -        | 30        | 70       | 100   |



**\*\*University elective courses: SCHEME OF TEACHING - Semester-II**

| Course Code | Course Name                         | Credit Hours | Credit Points | Hrs/Week |
|-------------|-------------------------------------|--------------|---------------|----------|
| CE771.01    | Project Management                  | 2            | 2             | 2        |
| ME781.01    | Occupational Health and Safety      | 2            | 2             | 2        |
| CA730       | Internet and Web Designing          | 2            | 2             | 2        |
| CA842.01    | Mobile Application Development      | 2            | 2             | 2        |
| PT796.01    | Fitness and Nutrition               | 2            | 2             | 2        |
| MB651       | Software Based Statistical Analysis | 2            | 2             | 2        |
| NR755       | First Aid and Life Support          | 2            | 2             | 2        |
| OC733.01    | Introduction to Polymer Science     | 2            | 2             | 2        |
| PH892.01    | Intellectual Property Rights        | 2            | 2             | 2        |
| PH893       | Clinical Data & Interpretation      | 2            | 2             | 2        |

**\*\*University elective courses: SCHEME OF EVALUATION- Semester-II**

| Course Code | Course Name                         | Evaluation Scheme |          |           |          |       |
|-------------|-------------------------------------|-------------------|----------|-----------|----------|-------|
|             |                                     | Theory            |          | Practical |          | Total |
|             |                                     | Internal          | External | Internal  | External |       |
| CE771.01    | Project Management                  | -                 | -        | 30        | 70       | 100   |
| ME781.01    | Occupational Health and Safety      | -                 | -        | 30        | 70       | 100   |
| CA730       | Internet and Web Designing          | -                 | -        | 30        | 70       | 100   |
| CA842.01    | Mobile Application Development      | -                 | -        | 30        | 70       | 100   |
| PT796.01    | Fitness and Nutrition               | -                 | -        | 30        | 70       | 100   |
| MB651       | Software Based Statistical Analysis | -                 | -        | 30        | 70       | 100   |
| NR755       | First Aid and Life Support          | -                 | -        | 30        | 70       | 100   |
| OC733.01    | Introduction to Polymer Science     | -                 | -        | 30        | 70       | 100   |
| PH892.01    | Intellectual Property Rights        | -                 | -        | 30        | 70       | 100   |
| PH893       | Clinical Data & Interpretation      | -                 | -        | 30        | 70       | 100   |

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**MASTER OF PHARMACY (PHARMACY PRACTICE) PROGRAMME**  
 Schemes for internal assessments and end semester examinations

**SEMESTER-III**

| Course Code  | Course Name                                       | Credit Hours | Credit Points | Hrs/Week  |
|--------------|---------------------------------------------------|--------------|---------------|-----------|
| PHCCC010     | Research Methodology and Biostatistics*           | 4            | 4             | 4         |
| PHCCC011     | Journal Club-I                                    | 1            | 1             | 1         |
| PHPPP016     | Discussion / Presentation (Proposal Presentation) | 2            | 2             | 2         |
| PHPPP017     | Research Work-I                                   | 28           | 14            | 28        |
| <b>Total</b> |                                                   | <b>35</b>    | <b>21</b>     | <b>35</b> |

\* Non University Exam

**SCHEME OF EVALUATION**

| Course Code | Course                                            | Internal Assessment |                 |          |       | End Semester Exams |          | Total Marks |
|-------------|---------------------------------------------------|---------------------|-----------------|----------|-------|--------------------|----------|-------------|
|             |                                                   | Continuous Mode     | Sessional Exams |          | Total | Marks              | Duration |             |
|             |                                                   |                     | Marks           | Duration |       |                    |          |             |
| PHCCC010    | Research Methodology and Biostatistics*           | 10                  | 15              | 1 Hrs    | 25    | 75                 | 3 Hrs    | 100         |
| PHCCC011    | Journal Club-I                                    | -                   | -               | -        | 25    | -                  | -        | 25          |
| PHPPP016    | Discussion / Presentation (Proposal Presentation) | -                   | -               |          | 50    | -                  |          | 50          |
| PHPPP017    | Research Work- I                                  | -                   | -               | -        | -     | 350                | 1 Hr     | 350         |
| Total       |                                                   |                     |                 |          |       |                    |          | 525         |

\* Non University Exam

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**MASTER OF PHARMACY (PHARMACY PRACTICE) PROGRAMME**  
Schemes for internal assessments and end semester examinations

**SEMESTER-IV**

| Course Code  | Course Name             | Credit Hours | Credit Points | Hrs/Week  |
|--------------|-------------------------|--------------|---------------|-----------|
| PHCCC012     | Journal Club-II         | 1            | 1             | 1         |
| PHPPP018     | Discussion/Presentation | 3            | 3             | 3         |
| PHPPP019     | Research work-II        | 31           | 16            | 31        |
| <b>Total</b> |                         | <b>35</b>    | <b>20</b>     | <b>35</b> |

**SCHEME OF EVALUATION**

| Course Code | Course                  | Internal Assessment |                 |              |           | End Semester Exams |              | Total Marks |
|-------------|-------------------------|---------------------|-----------------|--------------|-----------|--------------------|--------------|-------------|
|             |                         | Continuo<br>us Mode | Sessional Exams |              | Tota<br>l | Mark<br>s          | Duratio<br>n |             |
|             |                         |                     | Mark<br>s       | Duratio<br>n |           |                    |              |             |
| PHCCC012    | Journal Club-II         | -                   | -               | -            | 25        | -                  | -            | 25          |
| PHPPP018    | Discussion/Presentation | -                   | -               | -            | 75        | -                  | -            | 75          |
| PHPPP019    | Research work-II        | -                   | -               | -            | -         | 400                | I Hr         | 400         |
| Total       |                         |                     |                 |              |           |                    |              | 500         |

**SEMESTER WISE CREDITS DISTRIBUTION**

| Semester                                                                                                                                  | Credit Point               |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| I                                                                                                                                         | 25                         |
| II                                                                                                                                        | 27                         |
| III                                                                                                                                       | 21                         |
| IV                                                                                                                                        | 20                         |
| Co-curricular Activity<br>(Attending Conference, Scientific Presentations, Publication, Industry Training and other scholarly Activities) | Maximum 5                  |
| Total Credit Point                                                                                                                        | Minimum- 93<br>Maximum-98* |

\*Credit Point for co-curricular activity

**FACULTY OF PHARMACY**  
**Master of Pharmacy Programme**

**Syllabi**  
**Semester I**

**Charotar University of Science and Technology**

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP001: CLINICAL PHARMACY PRACTICE (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                  | <b>Minimum number of hours</b> |
|----------------|-------------------------------------------|--------------------------------|
| 1              | Introduction to Clinical Pharmacy         | 12                             |
| 2              | Clinical Pharmacy Services                | 12                             |
| 3              | Patient Data Analysis                     | 12                             |
| 4              | Lab Data Interpretation                   | 12                             |
| 5              | Medicines and Poison Information Services | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Introduction to Clinical Pharmacy</b>                                                                                                                                                                                                                                                                                                                                                                       | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Definition, evolution and scope of clinical pharmacy, International and national scenario of clinical pharmacy practice, pharmaceutical care</li> <li>Clinical Pharmacy Services: Ward round participation, Drug therapy review (drug therapy monitoring including medication order review, chart endorsement, clinical review and pharmacist interventions)</li> </ul> |              |                      |
| <b>2.</b>      | <b>Clinical Pharmacy Services</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Patient medication history interview, Basic concept of medicine and poison information services, Basic concept of pharmacovigilance, Hemovigilance, Materiovigilance and AEFI, Patient medication counselling, Drug utilization evaluation, Documentation of clinical pharmacy services, Quality assurance of clinical pharmacy services</li> </ul>                     |              |                      |
| <b>3.</b>      | <b>Patient Data Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Patient Data and Practice Skills: Patient's case history – its structure and significances in drug therapy management, Common medical abbreviations and terminologies used in clinical practice</li> <li>Communication skills: verbal and non-verbal communications, its applications in patient care services</li> </ul>                                               |              |                      |
| <b>4.</b>      | <b>Lab Data Interpretation</b>                                                                                                                                                                                                                                                                                                                                                                                 | <b>12</b>    | <b>20%</b>           |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |            |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|           | <ul style="list-style-type: none"> <li>Hematological tests, Renal function tests, Liver function tests, Tests associated with cardiac disorders, Pulmonary function tests, Thyroid function tests, Fluid and electrolyte balance, Microbiological culture sensitivity tests</li> </ul>                                                                                                                                                  |           |            |
| <b>5.</b> | <b>Medicines and Poison Information Services</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>12</b> | <b>20%</b> |
|           | <ul style="list-style-type: none"> <li>Medicine Information Service: Definition and need for medicine information service, Medicine information resources, Systematic approach in answering medicine information queries, Preparation of verbal and written response, Establishing a drug information centre.</li> <li>Poison Information Service: Definition, need, organization and functions of poison information centre</li> </ul> |           |            |

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------|
| CO1 | describe the elements of pharmaceutical care and provide comprehensive patient care services                    |
| CO2 | interpret the laboratory results to aid the clinical diagnosis of various disorders                             |
| CO3 | apply the basic knowledge and skills required to practice clinical pharmacy                                     |
| CO4 | identify medicine and poison information to enable healthcare professionals in the efficient patient management |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO2 | 3   | 3   | 3   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO3 | 3   | 3   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO4 | 3   | 3   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |

### ❖ References:

1. A Textbook of Clinical Pharmacy Practice – Essential concepts and skills – Parthasarathi G, Karin Nyfort-Hansen and Milap Nahata
2. Practice Standards and Definitions - The Society of Hospital Pharmacists of Australia
3. Basic skills in interpreting laboratory data - Scott LT, American Society of Health System Pharmacists Inc
4. Relevant review articles from recent medical and pharmaceutical literature.
5. Andrews EB, Moore N. Mann's Pharmacovigilance
6. Dipiro JT, Talbert RL, Yee GC. Pharmacotherapy: A Pathophysiologic Approach
7. Straus SE, Richardson WS, Glasziou P, Haynes RB. Evidence-Based Medicine: How to practice and teach it

8. Cohen MR. Medication Errors
9. Online:  
[http://medicinesaustralia.com.au/files/2012/05/MA\\_QUM\\_External\\_Reduced.pdf](http://medicinesaustralia.com.au/files/2012/05/MA_QUM_External_Reduced.pdf)  
<http://curriculum.racgp.org.au/statements/quality-use-of-medicines/>  
[http://www.rug.nl/research/portal/files/14051541/Chapter\\_2.pdf](http://www.rug.nl/research/portal/files/14051541/Chapter_2.pdf)
10. Relevant review articles from recent medical and pharmaceutical literature. 199

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP002: PHARMACOTHERAPEUTICS-I (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                           | <b>Minimum number of hours</b> |
|----------------|----------------------------------------------------|--------------------------------|
| 1              | Cardiovascular system                              | 12                             |
| 2              | Respiratory system                                 | 12                             |
| 3              | Gastrointestinal system                            | 12                             |
| 4              | Gastrointestinal system and Hematological diseases | 12                             |
| 5              | Bone and joint disorders                           | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                         | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Cardiovascular system</b>                                                                                                                                                                                                                                        | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Hypertension, Congestive cardiac failure, Acute coronary syndrome, Arrhythmias, Hyperlipidemias.</li> </ul>                                                                                                                  |              |                      |
| <b>2.</b>      | <b>Respiratory system</b>                                                                                                                                                                                                                                           | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Asthma, Chronic obstructive airways disease, Drug induced pulmonary diseases</li> <li>Endocrine system: Diabetes, Thyroid diseases</li> </ul>                                                                                |              |                      |
| <b>3.</b>      | <b>Gastrointestinal system</b>                                                                                                                                                                                                                                      | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Peptic ulcer diseases, Reflux esophagitis, Inflammatory bowel diseases, Jaundice and hepatitis</li> </ul>                                                                                                                    |              |                      |
| <b>4.</b>      | <b>Gastrointestinal system and Hematological diseases</b>                                                                                                                                                                                                           | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Cirrhosis, Diarrhea and Constipation, Drug-induced liver disease</li> <li>Anemia, Deep vein thrombosis, Drug Induced hematological disorders</li> </ul>                                                                      |              |                      |
| <b>5.</b>      | <b>Bone and joint disorders</b>                                                                                                                                                                                                                                     | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Rheumatoid arthritis, Osteoarthritis, Gout, Osteoporosis</li> <li>Dermatological Diseases: Psoriasis, Eczema and scabies, impetigo, drug induced skin disorders</li> <li>Ophthalmology: Conjunctivitis, Glaucoma.</li> </ul> |              |                      |



**Course Outcome (COs):**

At the end of the course, the students would be able to

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| CO1 | describe the rationale for drug therapy for management of various disease conditions                |
| CO2 | compare the clinical controversies in drug therapy and evidence based medicine                      |
| CO3 | create and design individualized therapeutic plans based on diagnosis                               |
| CO4 | identify the patient specific parameters relevant in initiating drug therapy and monitoring therapy |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 3   | 3   | -   | -   | 3   | 3   | -   | -   | -    | -    |
| CO2 | 3   | 3   | 3   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO3 | 3   | 3   | 3   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO4 | 3   | 3   | 3   | -   | -   | -   | -   | -   | -   | -    | -    |

**❖ References:**

1. Roger and Walker. Clinical Pharmacy and Therapeutics – Churchill Livingstone publication
2. Joseph T. Dipiro et al. Pharmacotherapy: A Pathophysiologic Approach- Appleton & Lange
3. Robins SL. Pathologic basis of disease -W.B. Saunders publication
4. Eric T. Herfindal. Clinical Pharmacy and Therapeutics- Williams and Wilkins Publication
5. Lloyd Young and Koda-Kimble MA Applied Therapeutics: The clinical Use of Drugs Lippincott Williams and Wilkins
6. Chisholm- Burns Wells Schwinghammer Malone and Joseph P Dipiro. Pharmacotherapy Principles and practice— McGraw Hill Publication
7. Carol Mattson Porth. Principles of Pathophysiology- Lippincott Williams and Wilkins
8. Harrison's. Principles of Internal Medicine - McGraw Hill
9. Relevant review articles from recent medical and pharmaceutical literature

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP003: HOSPITAL AND COMMUNITY PHARMACY (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>      | <b>Minimum number of hours</b> |
|----------------|-------------------------------|--------------------------------|
| 1              | Introduction to Hospitals     | 12                             |
| 2              | Hospital Formulary Guidelines | 12                             |
| 3              | Education and training        | 12                             |
|                | Community Pharmacy management |                                |
| 4              | Prescription                  | 12                             |
| 5              | Health Promotion              | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Introduction to Hospitals</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Definition, classification, organizational structure</li> <li>Hospital Pharmacy: Definition, Relationship of hospital pharmacy department with other departments, Organizational structure, legal requirements, work load statistics, Infrastructural requirements, Hospital Pharmacy Budget and Hospital Pharmacy management</li> <li>Hospital Drug Policy: Pharmacy and Therapeutics Committee, Infection Control committee, Research and Ethics Committee, Management of Medicines as per NABH</li> </ul> |              |                      |
| <b>2.</b>      | <b>Hospital Formulary Guidelines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Its development, Developing Therapeutic guidelines, Drug procurement process, and methods of Inventory control, Methods of Drug distribution, Intravenous admixtures, Hospital Waste Management</li> </ul>                                                                                                                                                                                                                                                                                                   |              |                      |
| <b>3.</b>      | <b>Education and training:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Training of technical staff, training and continuing education for pharmacists, Pharmacy students, Medical staff and students, Nursing staff and students, Formal and informal meetings and lectures, Drug and therapeutics newsletter.</li> </ul>                                                                                                                                                                                                                                                           |              |                      |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|           | <ul style="list-style-type: none"> <li>Community Pharmacy Practice: Definition, roles and responsibilities of community pharmacists, and their relationship with other health care providers.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                |           |            |
|           | <b>Community Pharmacy management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |            |
|           | <ul style="list-style-type: none"> <li>Legal requirements to start community pharmacy, site selection, lay out &amp; design, drug display, super drug store model, accounts and audits, Good dispensing practices, Different software &amp; databases used in community pharmacies. Entrepreneurship in community pharmacy.</li> </ul>                                                                                                                                                                                                                                  |           |            |
| <b>4.</b> | <b>Prescription</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |            |
|           | <ul style="list-style-type: none"> <li>Legal requirements and interpretation, prescription related problems. Responding to symptoms of minor ailments: Headache, pyrexia, menstrual pains, food and drug allergy</li> <li>OTC medication: Rational use of over-the-counter medications, Medication counseling and use of patient information leaflets</li> <li>Medication adherence: Definition, factors influencing adherence behavior, strategies to improve medication adherence Patient referrals to the doctors ADR monitoring in community pharmacies.</li> </ul> | <b>12</b> | <b>20%</b> |
| <b>5.</b> | <b>Health Promotion</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |            |
|           | <ul style="list-style-type: none"> <li>Definition and health promotion activities, family planning, Health screening services, first aid, prevention of communicable and noncommunicable diseases, smoking cessation. Child and mother care National Health Programs- Role of Community Pharmacist in Malaria and TB control Programs. Home Medicines review program – Definition, objectives, Guidelines, method and outcomes. Research in community pharmacy Practice</li> </ul>                                                                                      | <b>12</b> | <b>20%</b> |

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | describe the organizational structure and functioning of hospital pharmacy                                                                               |
| CO2 | summarise the hospital drug policy, role of various hospital committees and therapeutic guidelines.                                                      |
| CO3 | specify various practices for drug procurement, inventory control, drug distribution, intravenous admixtures and hospital waste management.              |
| CO4 | describe the role and responsibilities of community pharmacist, and the management of community pharmacy.                                                |
| CO5 | describe various concepts related to prescriptions, OTC medicines, rational use of medicines, medication adherence, ADR monitoring and health promotion. |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | -   | 3   | 3   | -   | -   | 3   | 3   | 3   | 3   | -    | 3    |
| CO2 | -   | 3   | 3   | -   | 3   | 3   | 3   | 3   | 3   | -    | 3    |
| CO3 | -   | 3   | 3   | -   | 3   | 3   | 3   | 3   | -   | 3    | 3    |
| CO4 | -   | 3   | 3   | -   | 3   | 3   | 3   | 3   | 3   | -    | 3    |
| CO5 | -   |     |     | 3   |     | 3   | 3   | 3   | 3   |      | 3    |

**❖ References:**

1. Hospital Pharmacy - Hassan WE. Lea and Febiger publication.
2. Textbook of hospital pharmacy - Allwood MC and Blackwell.
3. Avery's Drug Treatment, Adis International Limited.
4. Community Pharmacy Practice – Ramesh Adepu, BSP Publishers, Hyderabad
5. Remington Pharmaceutical Sciences.
6. A text book of Hospital Pharmacy by S.H.Merchant & Dr. J.S. Qadry. Revised by R.K.Goyal & R.K. Parikh.
7. WHO consultative group report. 4. R.P.S. Vol.2. Part –B; Pharmacy Practice section.
8. Handbook of pharmacy – health care. Edt. Robin J Harman. The Pharmaceutical press.
9. Advances in Pharmaceutical Sciences – By Bean and Backet, Academic Press Publication.
10. Pharmacy Practice for Technicians – By Den A. Ballington et al., 2nd Edition, New age Publishers.
11. Hospital & Clinical Pharmacy-S Balasubramanian, N. Narayanan. PharmaMed Press
12. Relevant review articles from recent medical and pharmaceutical literature.

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP014: CLINICAL RESEARCH (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                                        | <b>Minimum number of hours</b> |
|----------------|-----------------------------------------------------------------|--------------------------------|
| 1              | <b>Drug Development Process</b>                                 | 12                             |
| 2              | <b>Types and Designs used in Clinical Research</b>              | 12                             |
| 3              | <b>Clinical Trial Documents</b>                                 | 12                             |
| 4              | <b>Investigational Product</b>                                  | 12                             |
|                | <b>Clinical Trial Monitoring and Close out</b>                  |                                |
| 5              | <b>Quality Assurance and Quality Control in Clinical Trials</b> | 12                             |
|                | <b>Data Management</b>                                          |                                |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| 1.             | <b>Drug Development Process:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 12           | 20%                  |
|                | <ul style="list-style-type: none"> <li>• Introduction, various approaches to drug discovery, Investigational new drug application submission</li> <li>• Ethics in Biomedical Research: Ethical issues in biomedical research – Principles of ethics in biomedical research, Ethical committee [Institutional Review Board] - its constitution and functions, Challenges in implementation of ethical guidelines, ICH, GCP guidelines and ICMR guidelines in conduct of Clinical trials, Drug Safety Reporting</li> </ul>                                                                                                       |              |                      |
| 2.             | <b>Types and Designs used in Clinical Research</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 12           | 20%                  |
|                | <ul style="list-style-type: none"> <li>• Planning and execution of clinical trials, Various Phases of clinical trials, Bioavailability and Bioequivalence studies, Randomization techniques (Simple randomization, restricted randomization, blocking method and stratification), Types of research designs based on Controlling Method (Experimental, Quasi experimental, and Observational methods), Time Sequences (Prospective and Retrospective), Sampling methods (Cohort study, case Control study and cross sectional study), Health outcome measures (Clinical and Physiological, Humanistic and Economic)</li> </ul> |              |                      |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|           | <ul style="list-style-type: none"> <li>Clinical Trial Study team: Roles and responsibilities of: Investigator, Study Coordinator, Sponsor, Monitor, Contract Research Organization.</li> </ul>                                                                                                                                                                                                                                                                                                                                 |           |            |
| <b>3.</b> | <b>Clinical Trial Documents</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |            |
|           | <ul style="list-style-type: none"> <li>Guidelines to the preparation of following documents: Protocols, Investigator's Brochure, Informed Consent Form, Case report forms, Contracts and agreements, Dairy Cards</li> <li>Clinical Trial Start up activities: Site Feasibility Studies, Site/Investigator selection, Pre-study visit, Investigator meeting, Clinical trial agreement execution, Ethics committee document preparation and submission</li> </ul>                                                                | <b>12</b> | <b>20%</b> |
| <b>4.</b> | <b>Investigational Product</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |            |
|           | <ul style="list-style-type: none"> <li>Procurement and Storage of investigation product</li> <li>Filing procedures: Essential documents for clinical trial, Trial Master File preparation and maintenance, Investigator Site File, Pharmacy File, Site initiation visit, Conduct, Report and Follow up</li> </ul>                                                                                                                                                                                                              |           |            |
|           | <b>Clinical Trial Monitoring and Close out</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>12</b> | <b>20%</b> |
|           | <ul style="list-style-type: none"> <li>Preparation and conduct of monitoring visit: Review of source documents, CRF, ICF, IP storage, accountability and reconciliation, Study Procedure, EC communications, Safety reporting, Monitoring visit reporting and follow-up</li> <li>Close-Out visit: Study related documents collection, Archival requirement, Investigational Product reconciliation and destruction, Close-Out visit report.</li> </ul>                                                                         |           |            |
| <b>5.</b> | <b>Quality Assurance and Quality Control in Clinical Trials</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |            |
|           | <ul style="list-style-type: none"> <li>Types of audits, Audit criteria, Audit process, Responsibilities of stakeholders in audit process, Audit follow-up and documentation, Audit resolution and Preparing for FDA inspections, Fraud and misconduct management</li> </ul>                                                                                                                                                                                                                                                    |           |            |
|           | <b>Data Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>12</b> | <b>20%</b> |
|           | <ul style="list-style-type: none"> <li>Infrastructure and System Requirement for Data Management: Electronic data capture systems, Selection and implementation of new systems, System validation and test procedures, Coding dictionaries, Data migration and archival</li> <li>Clinical Trial Data Management: Standard Operating Procedures, Data management plan, CRF &amp; Data base design considerations, Study set-up, Data entry, CRF tracking and corrections, Data cleaning, Managing laboratory and ADR</li> </ul> |           |            |

|  |                                                                                                                   |  |  |
|--|-------------------------------------------------------------------------------------------------------------------|--|--|
|  | data, Data transfer and database lock, Quality Control and Quality Assurance in CDM, Data mining and warehousing. |  |  |
|--|-------------------------------------------------------------------------------------------------------------------|--|--|

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                         |
|-----|---------------------------------------------------------|
| CO1 | describe the process of drug development                |
| CO2 | Formulate and design clinical trial protocols           |
| CO3 | describe regulatory aspects related to clinical trials. |
| CO4 | preparation of documents related to clinical trial      |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 3   | 3   | 3   | -   |     | 3   | 3   |     | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   |     | 3   |     | 3   | -    | 3    |
| CO3 | 3   |     | 3   | -   |     | -   | 3   | 3   | 3   | -    | 3    |
| CO4 | 3   | 3   |     | -   | -   | 3   | 3   | 3   | 3   | -    | 3    |

### ❖ References

1. Principles and practice of pharmaceutical medicine, Second edition. Authors: Lionel. D. Edward, Aadrew.J.Fletcher Anthony W Fos , Peter D Sloaier Publisher: Wiley.
2. Handbook of clinical research. Julia Lloyd and Ann Raven Ed. Churchill Livingstone
3. Principles of Clinical Research edited by Giovanna di Ignazio, Di Giovanna and Haynes.
4. Central Drugs Standard Control Organization. Good Clinical Practices- Guidelines for Clinical Trials on Pharmaceutical Products in India. New Delhi: Ministry of Health.
5. International Conference on Harmonisation of Technical requirements for registration of Pharmaceuticals for human use. ICH Harmonised Tripartite Guideline. Guideline for Good Clinical Practice.E6; May 1996.
6. Ethical Guidelines for Biomedical Research on Human Subjects. Indian Council of Medical Research, New Delhi.
7. Textbook of Clinical Trials edited by David Machin, Simon Day and Sylvan Green, John Wiley and Sons.
8. Clinical Data Management edited by R K Rondels, S A Varley, C F Webbs. Second Edition, Jan 2000, Wiley Publications.
9. Goodman & Gilman: JG Hardman, LE Limbard, McGraw Hill Publications.
10. Relevant review articles from recent medical and pharmaceutical literature.

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP005: PHARMACY PRACTICE PRACTICAL-I (Practical)**

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**Total hours: 180 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the practical</b>                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------|
| 1              | Treatment Chart Review (one)                                                                                       |
| 2              | Medication History Interview (one)                                                                                 |
| 3              | Patient Medication Counseling (two)                                                                                |
| 4              | Drug Information Query (two)                                                                                       |
| 5              | Poison Information Query (one)                                                                                     |
| 6              | Lab Data Interpretation (two)                                                                                      |
| 7              | Presentation of clinical cases of various disease conditions adopting Pharmaceutical Care Plan Model (eight)       |
| 8              | ABC Analysis of a given list of medications (one)                                                                  |
| 9              | Preparation of content of a medicine, with proper justification, for the inclusion in the Hospital formulary (one) |
| 10             | Formulation and dispensing of a given IV admixtures (one)                                                          |
| 11             | Preparation of a patient information leaflet (two)                                                                 |
| 12             | Preparation of Study Protocol (one)                                                                                |
| 13             | Preparation of Informed Consent Form (one)                                                                         |

**Course Outcome (COs):**

At the end of the course, the students would be able to

|     |                                                            |
|-----|------------------------------------------------------------|
| CO1 | design the concept of drug treatment chart review          |
| CO2 | apply the pharmaceutical care plan for various diseases    |
| CO3 | formulate information related to medicine and poison query |
| CO4 | demonstrate the concept of patient counselling             |

**Course Articulation Matrix:**



|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | -   | -    | 3    |
| CO3 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | -   | -    | 3    |
| CO4 | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**HS105.02 B: ACADEMIC SPEAKING AND PRESENTATION SKILLS**  
**(Practical)**

**I. Credits and Schemes:**

| Se<br>m | Course<br>Code | Course<br>Name                                        | Credit<br>s | Teaching<br>Scheme        | Evaluation Scheme |              |              |              |           |
|---------|----------------|-------------------------------------------------------|-------------|---------------------------|-------------------|--------------|--------------|--------------|-----------|
|         |                |                                                       |             | Contact<br>Hours/We<br>ek | Theory            |              | Practical    |              | Tota<br>l |
|         |                |                                                       |             |                           | Intern<br>al      | Extern<br>al | Intern<br>al | Extern<br>al |           |
| 1       | HS105.02 B     | Academ<br>ic Speakin<br>g and Present<br>ation Skills | 02          | 02                        | --                | --           | 30           | 70           | 100       |

**II. Course Outline**

| Module<br>No. | Title/Topic                                                                                                                                                                                                                                                                                                                                                         | Classroom<br>Contact Hours |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1             | <b>Foundations of Advance Communication</b> <ul style="list-style-type: none"> <li>• Meaning and Definition of Advance Communication</li> <li>• Advance Communication in Digital, Social, Mobile World</li> <li>• Strategies for Advance Communication</li> <li>• Meaning and Concept of Academic Language</li> <li>• High Frequency Academic Vocabulary</li> </ul> | 04                         |
| 2             | <b>Art of Conversation</b> <ul style="list-style-type: none"> <li>• Describing people, places and things</li> <li>• Expressing opinions</li> <li>• Making suggesting</li> <li>• Persuading someone</li> <li>• Interpreting and Summarizing</li> </ul>                                                                                                               | 06                         |
| 3             | <b>Science of Power Speaking</b> <ul style="list-style-type: none"> <li>• Phonemes</li> <li>• Word Stress</li> <li>• Pronunciation</li> <li>• Intonation</li> <li>• Pause</li> <li>• Register</li> <li>• Fluency</li> </ul>                                                                                                                                         | 06                         |

|              |                                                                                                                                                                                                     |           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|              | <ul style="list-style-type: none"> <li>• Prosody</li> <li>• Lexical Range</li> </ul>                                                                                                                |           |
| <b>4</b>     | <b>Academic Speaking Application – Part I</b> <ul style="list-style-type: none"> <li>• Art of Oratory</li> <li>• Formal Presentation</li> <li>• Speech Analysis – Decoding Best Speeches</li> </ul> | <b>08</b> |
| <b>5</b>     | <b>Academic Speaking Application – Part II</b> <ul style="list-style-type: none"> <li>• Job Interview</li> <li>• Group Discussion</li> <li>• Meeting</li> </ul>                                     | <b>06</b> |
| <b>Total</b> |                                                                                                                                                                                                     | <b>30</b> |

### III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

### IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

#### Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

| Sl. No.      | Component                          | Number | Marks per incidence | Total Marks |
|--------------|------------------------------------|--------|---------------------|-------------|
| 1            | I-Talk                             | 1      | 10                  | 25          |
| 2            | Situational Speaking               | 1      | 05                  |             |
| 3            | Case Study - Speech Analysis       | 2      | 10                  |             |
| 4            | Attendance and Class Participation | -      |                     | 05          |
| <b>Total</b> |                                    |        |                     | <b>30</b>   |

#### External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

| Sl. No.      | Component        | Number | Marks per incidence | Total Marks |
|--------------|------------------|--------|---------------------|-------------|
| 1            | Viva / Practical | -      | 70                  | 70          |
| <b>Total</b> |                  |        |                     | <b>70</b>   |

**Course Outcome (COs):**

After completion of the course the student would be able to:

|     |                                                                                |
|-----|--------------------------------------------------------------------------------|
| CO1 | Understand and demonstrate advance communication skills and academic speaking. |
| CO2 | Demonstrate linguistic competence                                              |
| CO3 | Demonstrate performing ability at group discussion and personal interview.     |
| CO4 | Demonstrate the formal presentation skills.                                    |
| CO5 | Demonstrate ability to communicate in diverse situations                       |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO2 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO3 | -   | -   | -   | -   | 3   | -   | -   | 3   | -   | -    | -    |
| CO4 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO5 | -   | 2   | -   | -   | 3   | -   | -   | 3   | -   | -    | -    |

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

**V. Reference Books**

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

**VI. Reading**

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

**FACULTY OF PHARMACY**  
**Master of Pharmacy Programme**

**Syllabi**  
**Semester II**

**Charotar University of Science and Technology**

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP015: PRINCIPLES OF QUALITY USE OF MEDICINES(Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                              | <b>Minimum number of hours</b> |
|----------------|-------------------------------------------------------|--------------------------------|
| 1              | <b>Introduction to Quality use of medicines (QUM)</b> | 12                             |
| 2              | <b>Concepts in QUM</b>                                | 12                             |
| 3              | <b>QUM in Various Settings</b>                        | 12                             |
| 4              | <b>Regulatory Aspects of QUM in India</b>             | 12                             |
| 5              | <b>Medication Errors</b>                              | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Introduction to Quality use of medicines (QUM)</b>                                                                                                                                                                                                                                                                                                                                                                                                              | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Definition and Principles of QUM, Key partners and responsibilities of the partners, Building blocks in QMC, Evaluation process in QMC, Communication in QUM, Cost effective prescribing</li> </ul>                                                                                                                                                                                                                         |              |                      |
| <b>2.</b>      | <b>Concepts in QUM</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Evidence based medicine: Definition, concept of evidence-based medicine, Approach and practice of evidence based medicine in clinical settings</li> <li>Essential drugs: Definition, need, concept of essential drug, National essential drug policy and list</li> <li>Rational drug use: Definition, concept and need for rational drug use, Rational drug prescribing, Role of pharmacist in rational drug use</li> </ul> |              |                      |
| <b>3.</b>      | <b>QUM in Various Settings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Hospital settings, Ambulatory care/Residential care, Role of health care professionals in promoting the QUM, Strategies to promote the QUM, Impact of QUM on E-health, integrative medicine and multidisciplinary care.</li> <li>QUM in special population: Pediatric prescribing, Geriatric prescribing, Prescribing in pregnancy and lactation, Prescribing in immune compromised and organ failure patients.</li> </ul>  |              |                      |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |            |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
| <b>4.</b> | <b>Regulatory Aspects of QUM in India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>12</b> | <b>20%</b> |
|           | <ul style="list-style-type: none"> <li>Regulatory aspects of QUM in India: Regulation including scheduling, Regulation of complementary medicines, Regulation of OTC medicines, Professional responsibility of pharmacist, Role of industry in QUM in medicine development</li> </ul>                                                                                                                                                                                                                                             |           |            |
| <b>5.</b> | <b>Medication Errors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>12</b> | <b>20%</b> |
|           | <ul style="list-style-type: none"> <li>Definition, categorization and causes of medication errors, Detection and prevention of medication errors, Role of pharmacist in monitoring and management of medication errors</li> <li>Pharmacovigilance: Definition, aims and need for pharmacovigilance, Types, predisposing factors and mechanism of adverse drug reactions (ADRs), Detection, reporting and monitoring of ADRs, Causality assessment of ADRs, Management of ADRs, Role of pharmacist in pharmacovigilance</li> </ul> |           |            |

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                       |
|-----|-----------------------------------------------------------------------|
| CO1 | describe the quality use of medicines                                 |
| CO2 | describe concepts of quality use of medicine                          |
| CO3 | differentiate the role of quality use of medicine in various settings |
| CO4 | describe regulatory aspects of quality use of medicine                |
| CO5 | identify and resolve medication errors                                |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | 3   | -   | -   | -   | 3   | -   | 3   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | 3   | -    | 3    |
| CO3 | 3   | -   | -   | -   | -   | 3   | 3   | 3   | 3   | -    | 3    |
| CO4 | 3   | -   | 3   | -   | -   | -   | 3   | 3   | 3   | -    | 3    |
| CO5 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | 3   | -    | 3    |

### ❖ References:

1. A Textbook of Clinical Pharmacy Practice – Essential concepts and skills – Parthasarathi G, Karin Nyfort-Hansen and Milap Nahata
2. Andrews EB, Moore N. Mann's Pharmacovigilance
3. Dipiro JT, Talbert RL, Yee GC. Pharmacotherapy: A Pathophysiologic Approach

4. Straus SE, Richardson WS, Glasziou P, Haynes RB. Evidence-Based Medicine: How to practice and teach it
5. Cohen MR. Medication Errors
6. Online:  
[http://medicinesaustralia.com.au/files/2012/05/MA\\_QUM\\_External\\_Reduced.pdf](http://medicinesaustralia.com.au/files/2012/05/MA_QUM_External_Reduced.pdf)  
<http://curriculum.racgp.org.au/statements/quality-use-of-medicines/>  
[http://www.rug.nl/research/portal/files/14051541/Chapter\\_2.pdf](http://www.rug.nl/research/portal/files/14051541/Chapter_2.pdf)
7. Relevant review articles from recent medical and pharmaceutical literature.



**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP007: PHARMACOTHERAPEUTICS-II (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                        | <b>Minimum number of hours</b> |
|----------------|-------------------------------------------------|--------------------------------|
| 1              | Nervous system                                  | 12                             |
| 2              | Psychiatric disorders and Renal System          | 12                             |
| 3              | Infectious diseases                             | 12                             |
| 4              | Infectious diseases and Gynecological disorders | 12                             |
| 5              | Oncology                                        | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                           | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Nervous system</b>                                                                                                                                                                                                                                                 | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Epilepsy, Parkinson's disease, Stroke, Headache, Alzheimer's disease, Neuralgias and Pain pathways and Pain management.</li> </ul>                                                                                             |              |                      |
| <b>2.</b>      | <b>Psychiatric disorders and Renal System</b>                                                                                                                                                                                                                         | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Schizophrenia, Depression, Anxiety disorders, Sleep disorders, Drug induced psychiatric disorders</li> <li>Renal system: Acute renal failure, Chronic renal failure, Renal dialysis, Drug induced renal disease</li> </ul>     |              |                      |
| <b>3.</b>      | <b>Infectious diseases</b>                                                                                                                                                                                                                                            | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>General guidelines for the rational use of antibiotics and surgical prophylaxis, Urinary tract infections, Respiratory tract infections, Gastroenteritis, Tuberculosis, Malaria, Bacterial Endocarditis, Septicemia</li> </ul> |              |                      |
| <b>4.</b>      | <b>Infectious diseases and Gynecological disorders</b>                                                                                                                                                                                                                | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>Meningitis, HIV and opportunistic infections, Rheumatic fever, Dengue fever, H1N1, Helmenthiasis, Fungal infections.</li> <li>Gynecological disorders: Dysmenorrhea, Hormone replacement therapy.</li> </ul>                   |              |                      |
| <b>5.</b>      | <b>Oncology</b>                                                                                                                                                                                                                                                       | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>General principles of cancer chemotherapy, pharmacotherapy of breast cancer, lung cancer, head and</li> </ul>                                                                                                                  |              |                      |

|  |                                                                                             |  |  |
|--|---------------------------------------------------------------------------------------------|--|--|
|  | neck cancer, hematological malignancies, Management of nausea and vomiting, Palliative care |  |  |
|--|---------------------------------------------------------------------------------------------|--|--|

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| CO1 | describe the rationale for drug therapy for management of various disease conditions                |
| CO2 | compare the clinical controversies in drug therapy and evidence based medicine                      |
| CO3 | create and design individualized therapeutic plans based on diagnosis                               |
| CO4 | identify the patient specific parameters relevant in initiating drug therapy and monitoring therapy |
| CO5 | describe pathophysiology, clinical features, diagnosis and complications of various diseases.       |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 3   | -   | 3   | -   | 3   | 3   | -   | 3   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 3   | 3   | -   | 3   | -    | 3    |
| CO3 | 3   | 3   | -   | -   | -   | 3   | 3   | -   | 3   | -    | 3    |
| CO4 | 3   | 3   | 3   | -   | -   | 3   | 3   | 3   | 3   | -    | 3    |
| CO5 | 3   | -   | -   | -   | -   | 3   | -   |     | 3   |      | 3    |

### ❖ References:

1. Roger and Walker. Clinical Pharmacy and Therapeutics – Churchill Livingstone publication.
2. Joseph T. Dipiro et al. Pharmacotherapy: A Pathophysiologic Approach- Appleton & Lange
3. Robins SL. Pathologic basis of disease -W.B. Saunders publication Relevant review articles from recent medical and pharmaceutical literature.
4. Eric T. Herfindal. Clinical Pharmacy and Therapeutics- Williams and Wilkins Publication
5. Lloyd Young and Koda-Kimble MA Applied Therapeutics: The clinical Use of Drugs- Lippincott Williams and Wilkins
6. Chisholm- Burns Wells Schwinghammer Malone and Joseph P Dipiro. Pharmacotherapy Principles and practice— McGraw Hill Publication
7. Carol Mattson Porth. Principles of Pathophysiology- Lippincott Williams and Wilkins
8. Harrison's. Principles of Internal Medicine - McGraw Hill
9. Relevant review articles from recent medical and pharmaceutical literature

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP008: CLINICAL PHARMACOKINETICS AND THERAPEUTIC**  
**DRUG MONITORING (Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                  | <b>Minimum number of hours</b> |
|----------------|-------------------------------------------|--------------------------------|
| 1              | Introduction to Clinical pharmacokinetics | 12                             |
| 2              | Pharmacokinetics of Drug Interaction      | 12                             |
| 3              | Non Linier Mixed Effects Modelling        | 12                             |
| 4              | Altered Pharmacokinetics                  | 12                             |
| 5              | Therapeutic Drug monitoring               | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1</b>       | <b>Introduction to Clinical pharmacokinetics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>• Compartmental and Non compartmental models, Renal and non-renal clearance, Organ extraction and models of hepatic clearance, Estimation and determinants of bioavailability, Multiple dosing, Calculation of loading and maintenance doses</li> <li>• Designing of dosage regimens: Determination of dose and dosing intervals, Conversion from intravenous to oral dosing, Nomograms and Tabulations in designing dosage regimen</li> </ul>                                                                        |              |                      |
| <b>2</b>       | <b>Pharmacokinetics of Drug Interaction</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>• Pharmacokinetic drug interactions, Inhibition and Induction of Drug metabolism, Inhibition of Biliary Excretion</li> <li>• Pharmacogenetics: Genetic polymorphism in Drug metabolism: Cytochrome P-450 Isoenzymes, Genetic Polymorphism in Drug Transport and Drug Targets, Pharmacogenetics and Pharmacokinetic / Pharmacodynamic Considerations Introduction to Pharmacometrics: Introduction to Bayesian Theory, Adaptive method or Dosing with feedback, Analysis of Population pharmacokinetic Data</li> </ul> |              |                      |
| <b>3</b>       | <b>Non Linier Mixed Effects Modelling</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>12</b>    | <b>20%</b>           |
|                | <ul style="list-style-type: none"> <li>• The Structural or Base Model, Modeling Random Effects, Modeling Covariate Relationships, Mixture Model, estimation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                       |              |                      |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |            |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|          | Methods, Model Building Techniques, Covariate Screening Methods, Testing the model assumptions, Precision of the parameter estimates And confidence intervals, Model misspecification and violation of the model assumptions, Model Validation, Simulation of dosing regimens and dosing recommendations, Pharmacometrics software                                                                                                                                                                                                                                                                                                                          |           |            |
| <b>4</b> | <b>Altered Pharmacokinetics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>12</b> | <b>20%</b> |
|          | <ul style="list-style-type: none"> <li>Drug dosing in the elderly, Drug dosing in the paediatrics, Drug dosing in the obese patients, Drug dosing in the pregnancy and lactation, Drug dosing in the renal failure and extracorporeal removal of drugs, Drug dosing in the in hepatic failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |           |            |
| <b>5</b> | <b>Therapeutic Drug monitoring</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>12</b> | <b>20%</b> |
|          | <ul style="list-style-type: none"> <li>Introduction, Individualization of drug dosage regimen (Variability – Genetic, age, weight, disease and Interacting drugs), Indications for TDM, Protocol for TDM, Pharmacokinetic/Pharmacodynamic, Correlation in drug therapy, TDM of drugs used in the following conditions: Cardiovascular disease: Digoxin, Lidocaine, Amiodarone; Seizure disorders: Phenytoin, Carbamazepine, Sodium Valproate; Psychiatric conditions: Lithium, Fluoxetine, Amitriptyline; Organ transplantations: Cyclosporine; Cytotoxic Agents: Methotrexate, 5-FU, Cisplatin; Antibiotics: Vancomycin, Gentamicin, Meropenem.</li> </ul> |           |            |

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                                                                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | explain the basic concepts in biopharmaceutics, pharmacokinetics such as ADME, and application of pharmacokinetics                                 |
| CO2 | design the drug dosage regimen for individual patients, interpret and correlate the plasma drug concentrations with patient's therapeutic outcomes |
| CO3 | apply the concept of pharmacokinetic modelling for the given data using the principles of pharmacometrics                                          |
| CO4 | identify drug-drug interactions and positive therapeutic outcomes by conducting therapeutic drug monitoring for various drugs                      |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO 10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|------|
| CO1 | 3   | 3   | -   | 3   | -   | 3   | -   | -   | 3   | -     | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | 3   | -     | 3    |
| CO3 | 3   | 3   | -   | 3   | -   | 3   | -   | -   | 3   | -     | 3    |
| CO4 | 3   | 3   | 3   | 3   | -   | 3   | -   | -   | 3   | -     | 3    |

**REFERENCES:**

1. Leon Shargel, Susanna Wu-Pong, Andrew Yu. Applied Biopharmaceutics & Pharmacokinetics. New York: Mc Graw Hill.
2. Peter L. Bonate. Pharmacokinetic - Pharmacodynamic Modeling and Simulation. Springer Publications.
3. Michael E. Burton, Leslie M. Shaw, Jerome J. Schentag, William E. Evans. Applied Pharmacokinetics & Pharmacodynamics: Principles of Therapeutic Drug Monitoring. Ippincott Williams & Wilkins.
4. Steven How-Yan Wong, Irving Sunshine. Handbook of Analytical Therapeutic Drug Monitoring and Toxicology. CRC Press, USA.
5. Soraya Dhillon, Andrzej Kostrzewski. Clinical pharmacokinetics. 1<sup>st</sup> edition. London: Pharmaceutical Press.
6. Joseph T. Dipiro, William J. Spruill, William E. Wade, Robert A. Blouin and Jane M. Pruemer. Concepts in Clinical Pharmacokinetics. American Society of Health-System Pharmacists, USA.
7. Malcolm Rowland, Thomas N. Tozer. Clinical Pharmacokinetics and pharmacodynamics: concepts and applications. Ippincott Williams & Wilkins, USA.
8. Evans, Schentag, Jusko. Applied pharmacokinetics. American Society of Health system Pharmacists, USA.
9. Michael E. Winter. Basic Clinical Pharmacokinetics. Ippincott Williams & Wilkins, USA.
10. Milo Gibaldi. Biopharmaceutics and Clinical Pharmacokinetics. Pharma Book Syndicate, USA.
11. Dhillon and Kostrzewski. Clinical pharmacokinetics. Pharmaceutical Press, London.
12. John E. Murphy. Clinical Pharmacokinetics. 5th edition. US: American Society of Health- System Pharmacist, USA.
13. Relevant review articles from recent medical and pharmaceutical literature.

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP009: PHARMACOEPIDEMIOLOGY & PHARMACOECONOMICS**  
**(Theory)**

**Total hours: 60 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the unit</b>                                                                | <b>Minimum number of hours</b> |
|----------------|-----------------------------------------------------------------------------------------|--------------------------------|
| 1              | Introduction to Pharmacoepidemiology                                                    | 12                             |
| 2              | Pharmacoepidemiological Methods                                                         | 12                             |
| 3              | Introduction to Pharmacoeconomics:                                                      | 12                             |
| 4              | Pharmacoeconomics evaluations                                                           | 12                             |
| 5              | Definition, Steps involved, Applications, Advantages and disadvantages of the following | 12                             |

**Detailed Syllabus:**

| <b>Sr. No.</b> | <b>UNIT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Hours</b> | <b>Weightage (%)</b> |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|
| <b>1.</b>      | <b>Introduction to Pharmacoepidemiology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |              |                      |
|                | <ul style="list-style-type: none"> <li>Definition, Scope, Need, Aims and Applications</li> <li>Outcome measurement: Outcome measures</li> <li>Drug use measures: Monetary units, Number of prescriptions, units of drug dispensed, defined daily doses, prescribed daily doses, Diagnosis and Therapy surveys, Prevalence, Incidence rate, Monetary units, number of prescriptions, unit of drugs dispensed, defined daily doses and prescribed daily doses, medications adherence measurements.</li> <li>Concept of risk: Measurement of risk, Attributable risk and relative risk, Time- risk relationship and odds ratio</li> </ul> | <b>12</b>    | <b>20%</b>           |
| <b>2.</b>      | <b>Pharmacoepidemiological Methods</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |                      |
|                | <ul style="list-style-type: none"> <li>Qualitative models: Drug Utilization Review</li> <li>Quantitative models: case reports, case series, Cross sectional studies, Cohort and case control studies, Calculation of Odds' ratio, Meta-analysis models</li> <li>Drug effects study in populations: Spontaneous reporting, Prescription event monitoring, Post marketing surveillance, Record linkage systems, Applications of Pharmacoepidemiology</li> </ul>                                                                                                                                                                          | <b>12</b>    | <b>20%</b>           |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |            |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
| <b>3.</b> | <b>Introduction to Pharmacoeconomics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |            |
|           | <ul style="list-style-type: none"> <li>Definition, history of Pharmacoeconomics, Need of Pharmacoeconomic studies in Indian healthcare system. Cost categorization and resources for cost estimation: Direct costs, Indirect costs, Intangible costs.</li> <li>Outcomes and Measurements of Pharmacoeconomics: Types of outcomes, Clinical outcome, Economic outcomes, Humanistic outcomes. Quality Adjusted Life Years, Disability Adjusted Life Years, Incremental Cost Effective Ratio, Average Cost Effective Ratio. Person Time, Willingness to Pay, Time Trade Off and Discounting.</li> </ul> | <b>12</b> | <b>20%</b> |
| <b>4.</b> | <b>Pharmacoeconomics evaluations</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |            |
|           | <ul style="list-style-type: none"> <li>Definition, Steps involved, Applications, Advantages and disadvantages of the following Pharmacoeconomic models: Cost Minimization Analysis (CMA), Cost Benefit Analysis (CBA), Cost Effective Analysis (CEA), Cost Utility Analysis (CUA), Cost of Illness (COI), Cost Consequences Analysis (COA).</li> </ul>                                                                                                                                                                                                                                               | <b>12</b> | <b>20%</b> |
| <b>5.</b> | <b>Definition, Steps involved, Applications, Advantages and disadvantages of the following</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |            |
|           | <ul style="list-style-type: none"> <li>Health related quality of life (HRQOL): Definition, Need for measurement of HRQOL, Common HRQOL measures. Definition, Steps involved, Applications of the following: Decision Analysis and Decision tree, Sensitivity analysis, Markov Modeling, Software used in Pharmacoeconomic analysis, Applications of Pharmacoeconomics</li> </ul>                                                                                                                                                                                                                     | <b>12</b> | <b>20%</b> |

### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------|
| CO1 | explain the fundamental principles of Pharmacoeconomics.                                                        |
| CO2 | describe the various epidemiological methods and their applications.                                            |
| CO3 | explain the fundamental principles of Pharmacoeconomics and perform the key Pharmacoeconomics analysis methods. |
| CO4 | identify and determine relevant cost and consequences associated with pharmacy products and services.           |
| CO5 | apply the Pharmacoeconomic decision analysis methods in various pharmacy settings.                              |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | 3   | -   | 3   | 3   | 3   | 3    | 3    |
| CO2 | 3   | -   | -   | -   | 3   | -   | 3   | -   | -   | 3    | 3    |
| CO3 | 3   | -   | -   | -   | 3   | -   | 3   | 3   | 3   | 3    | 3    |
| CO4 | 3   | -   | -   | 3   | 3   | -   | 3   | 3   | 3   | 3    | 3    |
| CO5 | 3   | -   | -   | -   | 3   | -   |     |     |     | 3    |      |

**❖ References:**

1. Rascati K L. Essentials of Pharmacoeconomics, Woulters Kluwer Lippincott Williams & Wilkins, Philadelphia.
2. Thomas E Getzen. Health economics. Fundamentals and Flow of Funds. John Wiley & Sons, USA.
3. Andrew Briggs, Karl Claxton, Mark Sculpher. Decision Modelling for Health Economic Evaluation, Oxford University Press, London.
4. Michael Drummond, Mark Sculpher, George Torrence, Bernie O'Brien and Greg Stoddart. Methods for the Economic Evaluation of Health Care Programmes Oxford University Press, London.
5. George E Mackinnon III. Understanding health outcomes and pharmacoeconomics.
6. Graker, Dennis. Pharmacoeconomics and outcomes.
7. Walley, Pharmacoeconomics.
8. Pharmacoeconomic – ed. by Nowakowska – University of Medical Sciences, Poznan.
9. Relevant review articles from recent medical and pharmaceutical literature



**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHPPP010: PHARMACY PRACTICE PRACTICAL-II (Practical)**

**Total hours: 180 Hr**

**Outline of the course:**

| <b>Sr. No.</b> | <b>Title of the practical</b>                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------|
| 1              | Causality assessment of adverse drug reactions (two)                                                         |
| 2              | Detection and management of medication errors (three)                                                        |
| 3              | Rational use of medicines in special population (three)                                                      |
| 4              | Presentation of clinical cases of various disease conditions adopting Pharmaceutical Care Plan Model (eight) |
| 5              | Calculation of Bioavailability and Bioequivalence from the given data (two)                                  |
| 6              | Interpretation of Therapeutic Drug Monitoring reports of a given patient (two)                               |
| 7              | Calculation of various Pharmacoeconomic outcome analysis for the given data (two)                            |
| 8              | Practical regarding development of Case Record Form (CRF).                                                   |
| 9              | Case studies on Pharmacoepidemiological studies.                                                             |

**Course Outcome (COs):**

At the end of the course, the students would be able to

|     |                                                                        |
|-----|------------------------------------------------------------------------|
| CO1 | apply the concept of rational drug therapy                             |
| CO2 | create the pharmaceutical care plan for various diseases               |
| CO3 | identify the medication errors and ADRs reporting                      |
| CO4 | interpret results of pharmacoeconomics and Therapeutic drug monitoring |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 3   | 3   | 3   | -   | 3   | 3   | -   | -   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | 3   | 3   | -   | -   | -    | 3    |
| CO3 | 3   | 3   | 3   | 3   | -   | 3   | 3   | 3   | -   | -    | 3    |
| CO4 | 3   | 3   | 3   | 3   | -   | 3   | 3   | -   | -   | -    | 3    |

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**HS106.02 B : ACADEMIC WRITING (Practical)**

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**I. Credits and Schemes:**

| Sem | Course Code | Course Name      | Credits | Teaching Scheme    | Evaluation Scheme |          |           |          |       |
|-----|-------------|------------------|---------|--------------------|-------------------|----------|-----------|----------|-------|
|     |             |                  |         | Contact Hours/Week | Theory            |          | Practical |          | Total |
|     |             |                  |         |                    | Internal          | External | Internal  | External |       |
| II  | HS106.02 B  | Academic Writing | 02      | 02                 | --                | --       | 30        | 70       | 100   |

**II. Course Outline**

| Module No. | Title / Topic                                                                                                                                                                                                                                                                                                                             | Classroom Contact Hours |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1          | <b>Academic Writing and Research Process</b> <ul style="list-style-type: none"> <li>• Introduction to Academic Writing</li> <li>• Academic Writing as a Part of Research</li> <li>• Types of Academic Writing</li> <li>• Features of Academic Writing</li> <li>• Importance of Good Academic Writing in various Academic Works</li> </ul> | 05                      |
| 2          | <b>Anatomy of Academic Writing</b> <ul style="list-style-type: none"> <li>• Academic Vocabulary</li> <li>• Simple and Complex Sentences</li> <li>• Organizing Paragraphs</li> <li>• The Writing Process</li> <li>• Adopting Academic Writing Style</li> </ul>                                                                             | 05                      |
| 3          | <b>Key Academic Skills</b> <ul style="list-style-type: none"> <li>• Note – taking</li> <li>• Note – making</li> <li>• Paraphrasing</li> <li>• Summarizing</li> </ul>                                                                                                                                                                      | 05                      |
| 4          | <b>Accuracy in Academic Writing</b> <ul style="list-style-type: none"> <li>• Lexical Range</li> <li>• Academic Language and Structures</li> <li>• Elements of Writing</li> <li>• Proof Reading, Editing, and Rewriting</li> </ul>                                                                                                         | 05                      |
| 5          | <b>Using and Citing Sources of Ideas</b> <ul style="list-style-type: none"> <li>• Academic Texts and their Types</li> <li>• Intellectual Honesty in Academic Writing</li> <li>• Avoiding Plagiarism – Idea Theft</li> </ul>                                                                                                               | 05                      |

|              |                                                                                                                                                                                                                                                           |           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|              | <ul style="list-style-type: none"> <li>Degrees of Plagiarism</li> <li>Types of Borrowing</li> <li>Anatomy of Citations</li> <li>Common Citation Styles</li> </ul>                                                                                         |           |
| 6            | <b>Contemporary Practices in Academic Writing</b> <ul style="list-style-type: none"> <li>Analytical Essays</li> <li>Graph / Table / Process Interpretation and Description</li> <li>Writing Reports</li> <li>Writing Research / Concept Papers</li> </ul> | 05        |
| <b>Total</b> |                                                                                                                                                                                                                                                           | <b>30</b> |

### III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

### IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

#### Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

| Sl. No. | Component                          | Number | Marks per incidence | Total Marks |
|---------|------------------------------------|--------|---------------------|-------------|
| 1       | Paragraph Writing                  | 1      | 3                   | 03          |
| 2       | Note-taking / Note-making          | 1      | 3                   | 03          |
| 3       | Paraphrasing / Summarizing         | 1      | 4                   | 04          |
| 4       | Essay Writing                      | 1      | 5                   | 05          |
| 5       | Concept Paper Writing              | 1      | 10                  | 10          |
| 5       | Attendance and Class Participation |        |                     | 05          |
| Total   |                                    |        |                     | 30          |

#### External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

| Sl. No.      | Component                                          | Number | Marks per incidence | Total Marks |
|--------------|----------------------------------------------------|--------|---------------------|-------------|
| 1            | Viva / Practical /Quiz/ Project / Academic Writing | -      | 70                  | 70          |
| <b>Total</b> |                                                    |        |                     | <b>70</b>   |

### Course Outcome (COs):

After completion of the course, the student would:

|     |                                                                                                          |
|-----|----------------------------------------------------------------------------------------------------------|
| CO1 | Have sound understanding of the concept and applications of academic writing                             |
| CO2 | Have acquired enough knowledge of academic writing style, strategy and approach                          |
| CO3 | Be able to demonstrate error free and effective academic writing                                         |
| CO4 | Be able to demonstrate ability to work on project/report/paper writing                                   |
| CO5 | Understand the concept of plagiarism and learn to use different citation styles as a part of referencing |

### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO2 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO3 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO4 | -   | -   | -   | -   | -   | -   | -   | 3   | -   | -    | -    |
| CO5 | -   | -   | -   | -   | -   | -   | 2   | -   | -   | -    | -    |

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

### V. Reference Books / Reading

#### Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

#### Essential Reading for Activity and Teacher Resource

- *Effective Academic Writing Level - 1,2,3,4 (Second Edition)* By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

#### Additional Reading

- Writing Your Thesis (2<sup>nd</sup> Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- [www.owl.perdue.edu](http://www.owl.perdue.edu)

**FACULTY OF PHARMACY**  
**Master of Pharmacy Programme**

**Syllabi**  
**Semester III**

**Charotar University of Science and Technology**

**FACULTY OF PHARMACY**  
**MASTER OF PHARMACY PROGRAMME**  
**PHCCC010: RESEARCH METHODOLOGY & BIostatISTICS (Theory)**  
**Total hours: 60 Hr**

**Outline of the course:**

| Sr. No. | Title of the unit                                | Minimum number of hours |
|---------|--------------------------------------------------|-------------------------|
| 1       | General Research Methodology                     | 12                      |
| 2       | Biostatistics                                    | 12                      |
| 3       | Medical Research                                 | 12                      |
| 4       | CPCSEA guidelines for laboratory animal facility | 12                      |
| 5       | Declaration of Helsinki                          | 12                      |

**Detailed Syllabus:**

| Sr. No. | UNIT                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hours | Weightage (%) |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------|
| 1.      | <b>General Research Methodology</b>                                                                                                                                                                                                                                                                                                                                                                                                   | 12    | 20%           |
|         | Research, objective, requirements, practical difficulties, review of literature, study design, types of studies, strategies to eliminate errors/bias, controls, randomization, crossover design, placebo, blinding techniques                                                                                                                                                                                                         |       |               |
| 2.      | <b>Biostatistics</b>                                                                                                                                                                                                                                                                                                                                                                                                                  | 12    | 20%           |
|         | Definition, application, sample size, importance of sample size, factors influencing sample size, dropouts, statistical tests of significance, type of significance tests, parametric tests(students “t” test, ANOVA, Correlation coefficient, regression), non-parametric tests (wilcoxon rank tests, analysis of variance, correlation, chi square test), null hypothesis, P values, degree of freedom, interpretation of P values. |       |               |
| 3.      | <b>Medical Research</b>                                                                                                                                                                                                                                                                                                                                                                                                               | 12    | 20%           |
|         | History, values in medical ethics, autonomy, beneficence, non-maleficence, double effect, conflicts between autonomy and beneficence/non-maleficence, euthanasia, informed consent,                                                                                                                                                                                                                                                   |       |               |

|           |                                                                                                                                                                                                                                                                                                                          |           |            |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|
|           | confidentiality, criticisms of orthodox medical ethics, importance of communication, control resolution, guidelines, ethics committees, cultural concerns, truth telling, online business practices, conflicts of interest, referral, vendor relationships, treatment of family members, sexual relationships, fatality. |           |            |
| <b>4.</b> | <b>CPCSEA guidelines for laboratory animal facility</b>                                                                                                                                                                                                                                                                  | <b>12</b> | <b>20%</b> |
|           | Goals, veterinary care, quarantine, surveillance, diagnosis, treatment and control of disease, personal hygiene, location of animal facilities to laboratories, anesthesia, euthanasia, physical facilities, environment, animal husbandry, record keeping, SOPs, personnel and training, transport of lab animals.      |           |            |
| <b>5.</b> | <b>Declaration of Helsinki</b>                                                                                                                                                                                                                                                                                           | <b>12</b> | <b>20%</b> |
|           | History, introduction, basic principles for all medical research, and additional principles for medical research combined with medical care.                                                                                                                                                                             |           |            |

#### Course Outcome (COs):

At the end of the course, the students would be able to

|     |                                                                           |
|-----|---------------------------------------------------------------------------|
| CO1 | narrate hierarchy of continue research by proper fundamental methodology. |
| CO2 | summarize the guidelines and ethical values in medical research.          |
| CO3 | prepare protocol for Animal study.                                        |
| CO4 | apply the concept of design of experiments in pharmaceutical research.    |

#### Course Articulation Matrix:

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO 10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -     | 3    |
| CO2 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -     | 3    |
| CO3 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -     | 3    |
| CO4 | 3   | -   | -   | -   | -   | -   | -   | 3   | -   | -     | 3    |

### **Recommended Study Material:**

#### **❖ References:**

1. Research In Education- John V. Best, John V. Kahn 7th edition, Published by: Phi Learning Pvt. Ltd
2. Research Methodology: Methods and Techniques. C.R. Kothari and Gaurav Garg, New Age International Publications.
3. Essentials of Research Methodology and Dissertation Writing. Kanan Yelikar, Jaypee Publishers
4. ICMR Ethical Guidelines for Biomedical Research  
([http://icmr.nic.in/ethical\\_guidelines.pdf](http://icmr.nic.in/ethical_guidelines.pdf))
5. A review of —Scientist in legal Systems, Journal of Forensic Sciences (JOFS),21(2),1976.
6. Donald Menzel, Jones, Howard Mumford; Boyd, Lyle G., Writing a technical paper, J. Chem. Edu., 1962, 39 (6), p A500.
7. Mitchell, K. and Glover, T. 2001. Introduction to Biostatistics. McGraw Hill, Publishing Co.



**FACULTY OF PHARMACY  
MASTER OF PHARMACY PROGRAMME  
PHCCC011: JOURNAL CLUB -I**

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**Total hours: 15**

**Course Outcome (COs):**

At the end of the course, the students would be able to

|            | <b>CO</b>                                               |
|------------|---------------------------------------------------------|
| <b>CO1</b> | present scientific literature and interpret the finding |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | 3   | 3   | -   | -   | -   | 3   | -   | -    | 3    |

**FACULTY OF PHARMACY  
MASTER OF PHARMACY PROGRAMME  
PHPPP017: RESEARCH WORK**

**Total hours: 420**

**Course Outcome (COs):**

At the end of the course, the students would be able to

|            |                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | define and describe research problem.                                                                                                                                 |
| <b>CO2</b> | illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation. |
| <b>CO3</b> | present a dissertation report integrating appropriate written and verbal communicative skills.                                                                        |
| <b>CO4</b> | efficiently use communication and information technology tools.                                                                                                       |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |
| CO3 | 3   | -   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |
| CO4 | 3   | -   | -   | -   | -   | -   | 3   | 3   | -   | -    | 3    |

**FACULTY OF PHARMACY**  
**Master of Pharmacy Programme**

**Syllabi**  
**Semester IV**

**Charotar University of Science and Technology**

**FACULTY OF PHARMACY  
MASTER OF PHARMACY PROGRAMME  
PHCCC12: JOURNAL CLUB -II**

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**Total hours: 15**

**Course Outcome (COs):**

At the end of the course, the students would be able to

|            | <b>CO</b>                                               |
|------------|---------------------------------------------------------|
| <b>CO1</b> | present scientific literature and interpret the finding |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | 3   | 3   | -   | -   | -   | 3   | -   | -    | 3    |

**FACULTY OF PHARMACY  
MASTER OF PHARMACY PROGRAMME  
PHPPP019: RESEARCH WORK-II**

**Total hours: 465**

**Course Outcome (COs):**

At the end of the course, the students would be able to

|            |                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | define and describe research problem.                                                                                                                                 |
| <b>CO2</b> | illustrate project management skills such as project design, scientific information and literature access, project implementation, data analysis, and interpretation. |
| <b>CO3</b> | present a dissertation report integrating appropriate written and verbal communicative skills.                                                                        |
| <b>CO4</b> | efficiently use communication and information technology tools.                                                                                                       |

**Course Articulation Matrix:**

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | 3    |
| CO2 | 3   | 3   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |
| CO3 | 3   | -   | 3   | 3   | -   | -   | -   | -   | -   | -    | 3    |
| CO4 | 3   | -   | -   | -   | -   | -   | 3   | 3   | -   | -    | 3    |